



LOGITECH FLOW

TECHNICAL WHITE-PAPER

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ABOUT THIS DOCUMENT

This Logitech Flow white-paper is intended for customers and users that want to better understand how Logitech Flow works and understand more about the technologies that drive it. Here you'll learn about the amazing features of Logitech Flow, including details on how easy to use and secure it is. Our goal is to provide the details about how this novel Logitech technology works, helping stakeholders better understand the it to ultimately make an informed decision.

INTRODUCTION TO LOGITECH FLOW

Using two or three computers at the same time is no problem with Logitech Flow. With Flow users can automatically switch between computers just by moving your cursor to the edge of the screen. It also makes transferring text, images and files between your computers effortless - just copy on one and paste on the other. Logitech Flow is extremely easy to setup and use. With a supported keyboard and the feature enabled, the system that has the mouse can also has the keyboard focus.

LOGITECH FLOW FEATURES

Logitech Flow provides multiple benefits to users, bringing multi-computer use to a new level. It's main features are:

- **Cursor Flow:** the ability for the mouse to move between computers when the user reaches the edge of the screen.
- **Copy and Paste:** the ability for the user to copy and paste text, images and files from computer to computer.
- **Keyboard Link:** the ability for the keyboard to follow the mouse cursor, switching to the same computer whenever the mouse switches.
- **Easy & Secure:** setting up Logitech Flow is extremely easy. Auto-discovery finds and configures your computers in seconds and all the data is transferred on a secure connection, with SSL and AES-256 bit encryption.

WHAT YOU NEED?

Logitech Flow is a feature available for select Logitech mice.

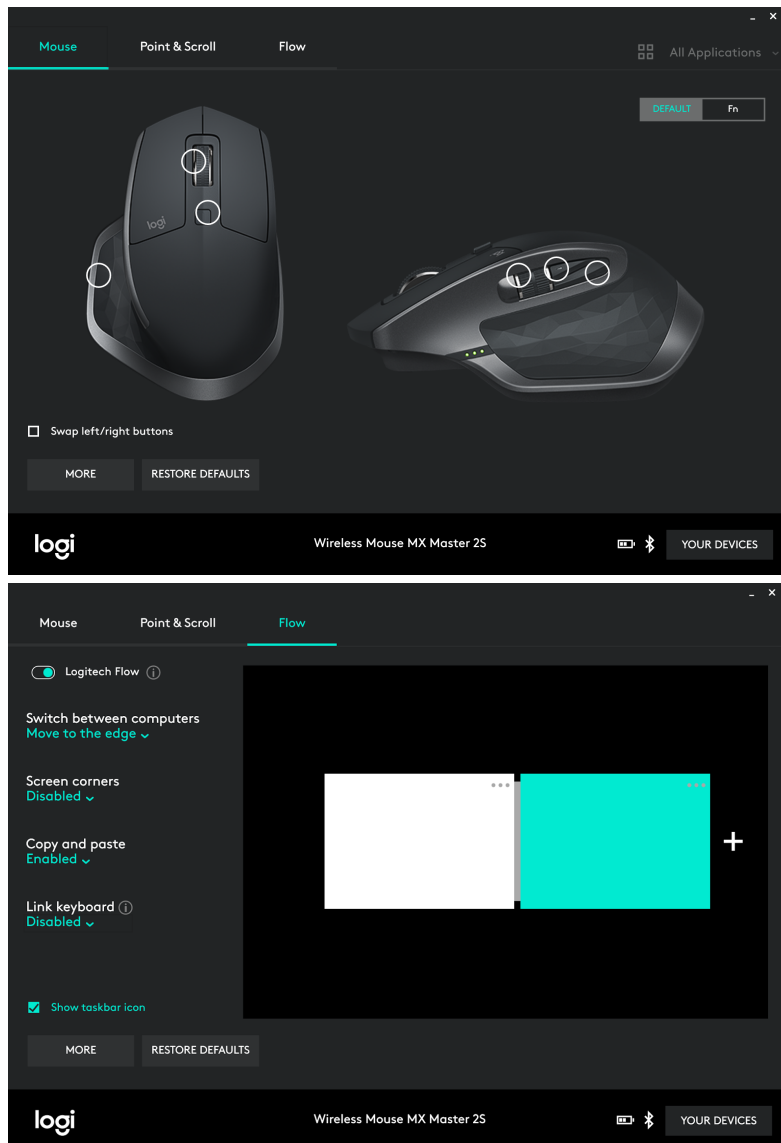
- You don't need a supported keyboard, although one is recommended
- A list of compatible devices can be found [here](#).

UNDERSTANDING LOGITECH FLOW

Several technological components make up Logitech Flow, below you'll find more information on the different components and their behavior.

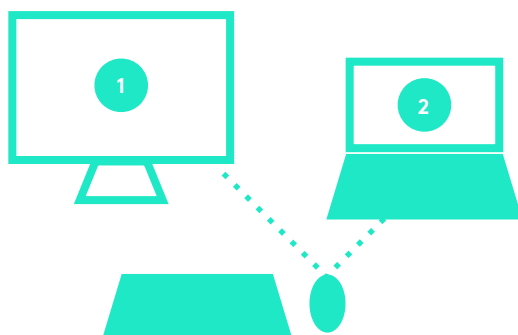
LOGITECH OPTIONS

Logitech Flow is a feature available on select devices, and is part of the Logitech Options application. For more information on Logitech Options and a list of supported devices, please visit the [Logitech Options website](#).



CURSOR FLOW

Using multiple computers at the same time with your favorite mouse and keyboard is a breeze with Logitech Flow. You'll be able to automatically switch between computers, Windows and macOS, just by moving your cursor to the edge of the screen.



EASY-SWITCH SIMPLIFIED DIAGRAM

Flow's main feature is its ability to switch the cursor control from one screen to the next, we call this Cursor Flow.

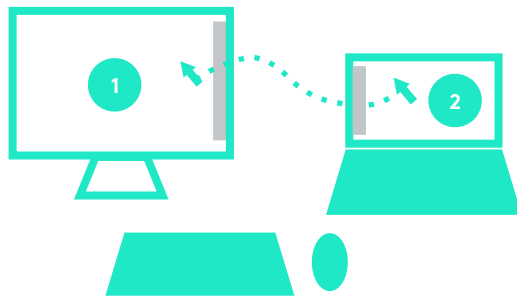
Cursor Flow relies on another Logitech technology called Easy-Switch. With Easy-Switch a mouse or keyboard can be paired to more than one device, allowing the user to switch between devices with the press of a button. Logitech Flow builds upon Easy-Switch, making the switch process more

EASY-SWITCH

Easy-Switch is the backbone of Logitech Flow

- Allows a mouse or keyboard to be connected to multiple computers
- Provides a reliable connection, much better than transferring cursor movement over the network

intuitive for users. When users reaches the edge of the screen, as per the arrangement of the computers in Logitech Options, the application signals the mouse to perform the Easy-Switch operation, thereby connecting the mouse to the neighbor computer.



EDGE DETECTION SIMPLIFIED DIAGRAM

switch when the Ctrl key on their keyboard is down. From Logitech Options, users can also assign the Ctrl key to a mouse button.

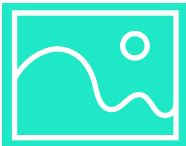
In order to provide a more seamless experience for users, and give the impression that the mouse cursor is the same on both computers, Flow transfers over the local network the position of the mouse on the first computer and repositions the cursor on the neighbor computer on a similar position - this information is transferred securely over the network (more details on the Flow Security section of this document). For users that want to avoid unintentional switches, users have the ability of only

EDGE DETECTION

When the mouse cursor reaches the edge of the screen Flow triggers Easy-Switch on the mouse (and keyboard)

- Flow repositions the cursor in the y-coordinate of the receiving computer's screen, so it feels like the same cursor is moving from one computer to the other.

COPY AND PASTE



TEXT, IMAGES, FILES & FOLDERS

Logitech Flow makes transferring information between your computers a wonderful experience. No more sending emails to yourself with attachments, just copy from one and paste on the other. It's as easy as it gets.

Logitech Flow allows users of both Windows and macOS machines to quickly and painlessly copy and paste text, both plain text and rich text, images, files and folders between peer computers. It achieves this by sending clipboard hints from the flow out computer to the flow in computer during channel switch. These hints contain enough information to allow the flow in computer to setup clipboard proxies. During a paste operation, the OS queries the proxy for the actual data to be pasted. Flow retrieves the data necessary to complete the paste from the flow out peer and hands it to the OS which then gives it to the pasting application. At no time is the data (including hints) stored in the cloud or otherwise sent to a third party computer. In fact the data never leaves the user's local network.

In keeping with the theme that all communications between hosts need to be protected from prying eyes (this is especially necessary when dealing with potentially sensitive user data), copy and paste packets are encrypted using the same ephemeral shared keys that are established during peer discovery and handshake. Again, Logitech Flow uses AES-256-CBC with random initialization vector generation for every packet to counter BEAST - a tool that exploits a vulnerability in TLS (Transport Layer Security) protocol 1.0 - and Message Authentication Codes (MAC) after encryption, to prevent the Cryptographic Doom Principle. Because the keys used to encrypt the packets are ephemeral, even if an adversary captures and stores packets across the wire, they will not be able to ever decrypt them as the keys are rotated every session and are not stored in permanent storage. This technique, known as Perfect Forward Secrecy, greatly reduces the risk from a 'man in the middle' attack (ability to listen in on the user's communications).

COPY & PASTE DATA

Flow transfers clipboard content between computers

- Users can use the commands (Ctrl-C, Ctrl-V, Cmd-C, Cmd-V) they're used to already
- All data is transferred securely over the local network, not over the internet.

KEYBOARD LINK



THE KEYBOARD FOLLOW THE MOUSE

Along with sharing mice between linked computers, Logitech Flow also allows the user to share a single Flow Compatible keyboard. On the Flow setup page, if Options determines that a Flow compatible keyboard is paired to both computers, it will give the user an option to allow Flow to switch the keyboard along with the mouse whenever a switch happens. Because mice and keyboards can be paired on different channels, Flow keeps track of what channel the keyboard was paired on, and perform the appropriate switch automatically.

With a compatible Logitech keyboard, you can have the best Flow experience. As you move the cursor to the other computer, your Logitech keyboard will connect directly to it. Focus on what you're typing, not where you're typing it on.

Along with sharing mice between linked computers, Logitech Flow also allows the user to

FLOW KEYBOARDS

Complete the Logitech Flow experience with a compatible keyboard.

- Keyboard follows the mouse when it switches
- A list of Flow compatible keyboards is available [here](#).

SET-UP PROCESS

We've designed Logitech Flow to not only be simple to use, but also simple to setup. Upon first launch, users will be presented with a tutorial that guides them through the setup process. Furthermore, users won't have to manually type IP addresses from one computer on the other. Either with a broadcast discovery method and/or a cloud assisted method, which relies on the Logitech Presence Service, Flow automatically exchanges all necessary information between computers to enable its functionality, easing the setup process for users.



THREE SIMPLE STEPS TO SET-UP FLOW

STEPS TO SET-UP

Flow was designed with ease of use in mind, to set-up Flow you need to, in all computers:

- Download & Install Logitech Options
- Pair the mouse
- Connect to the same network

COMPUTER DISCOVERY PROCESS



DISCOVERY PROCESS SIMPLIFIED DIAGRAM

With two computers connected to the same network, Flow uses different network discovery mechanisms to ensure both computers can talk to each other. After installing Logitech Options, the user is presented with the choice to enable Flow. If the user chooses to do so, the system enters the discovery state. During this state the system uses two different methods to find compatible neighbors. A neighboring system is said to be Flow compatible if the user paired the same mouse and/or keyboard to it.

The first discovery method, broadcast discovery, uses UDP broadcasts on the

local subnet. For majority of home users with a single subnet, this discovery method will suffice to discover the neighboring peers. In case Flow peer computers cannot be found using this method, Flow uses a second discovery method, which is cloud-assisted and relies on the Logitech Presence Service (LPS). More details on these discovery methods below.

DISCOVERY METHODS

Flow relies on either of two discovery methods

- The first is a broadcast that happens completely over the local network
- The second, is a cloud assisted method, which relies on the Logitech Presence Service.

BROADCAST DISCOVERY

Logitech Flow creates a peer-to-peer network between Flow enabled computers, this means every peer needs to perform it's own peer discovery. For that reason when a discovery is initiated Flow broadcasts a discovery packet to the local subnet. This packet contains enough information to allow the peers to initiate their own discovery procedures. Since every packet on the wire is encrypted, the only way a peer can respond to this packet is if the same device was paired to it.

A peer responds to the discovery packet by broadcasting beacons packets to advertise it's presence to other peers. Upon receipt of the a beacon packet its sender is said to be discovered by the peer. To acknowledge the discovery, the receiver of the beacon packet unicasts a ping packet to the beacon initiator. Once this ping packet is received the sender has discovered the peer (mutual discovery).

CLOUD ASSISTED DISCOVERY

For set-ups where the network topology is more segmented and whose peers are on different subnets, need to go through firewalls, and/or routers, to communicate with each other, Flow uses a cloud assisted discovery method. For this secondary discovery method, Flow relies on the Logitech's Presence Service (LPS).

With this system, users don't need to find the IP address of one computer and manually copy it to the other(s), during the first set-up process and if/when the computer is assigned another IP address. This secondary discovery method, is used only when computers can't find each other over the local network with Flow's primary local broadcast host discovery method.es all unused entries.

During initialization a host advertises its presence by contacting the LPS and registering it's IP addresses. These addresses are retrieved during discovery to mimic the broadcast discovery protocol described above. The only difference between the two discovery methods is the fact that LPS discovery simulates the broadcast of discovery packets and beacon packets by unicasting them to every peer on the retrieved ip list. Below, you'll find a timing diagram of the LPS discovery protocol.

During this process, each Flow enabled computer sends the following information to the LPS:

- **DeviceID:** a general device ID number, which is the same for each device model (e.g.: every MX Master 2S has a common DeviceID).
- **Device Serial Number:** this is unique for each device and allows us match computers that are paired to each device.
- **Device Channel:** this is the channel on the device that the computer is paired to, this allows Flow to 'switch' to the correct channel.
- **Host IP addresses:** this is/are the local IP address(es) of the computer, which allows Flow to finalize the discovery process.

NOTE: LPS clears its cache and purges all unused entries after 48hrs if the information is not updated by Flow enabled computers during this time (e.g.: Flow was disabled, there's no network connection, or the computer is off).

CONNECTION INITIATION

Either discovery method leads to the identification of Flow network peers. Once the discovery is completed, only the peers identified will be allowed to rejoin the Flow network in the future. This security feature prevents the hijacking of the Flow network by a rogue host.

The discovery protocol described above leads to the initiation of a secure TCP tunnel between every pair of peers. The initiator of this connection is the host with the lower device channel number. This means that the host with channel 0 will initiate connections to the hosts with channels 1 and 2. The host with channel 1 will initiate connection to the host with channel 2, and the host with channel 2 will not initiate any connections. This algorithm leads to a fully connected peer-to-peer network between every pair of hosts.

CONNECTION LIFETIME

The nature of a peer-to-peer network dictates that peers can come and go at random times and in random order. These connections need to be monitored for health and torn down when it is necessary to reclaim precious system resources. To achieve that, a simple heartbeat protocol is used to monitor the health of the connection. During idle time, every few seconds, a heartbeat packet is exchanged between every pair of networked peers. The absence of this packet indicates to the host that its peer has vanished and that it needs to initiate a connection shutdown. When the peer comes back, a new discovery and connection phase allows it to rejoin the peer-to-peer network.

LOGITECH FLOW SECURITY

Flow is designed, from the ground up, with security in mind. Below you can find more details on the security model utilized by Logitech Flow.

Flow's security mode uses a novel shared key protocol to bootstrap the secure tunnel. The basic security level starts by generating a shared key from the device id and serial number (newer devices will have a 256 bit cryptographically secure random string to add significantly more entropy available to the keying material). This key is used to begin the handshake. During the handshake, the systems negotiate a significantly stronger shared key using Diffie-Hellman and an 8Kb randomly generated, cryptographically secure public key.

This method has a number of desirable properties. It achieves a high level of security without requiring the user to do anything else.

CONCLUSION

Logitech Flow is an innovative technology that has been developed with ease of use and security in mind. By giving users control of multiple computers the ability to use the same mouse and keyboard on multiple system and share content between computers in a easy and secure way, Logitech Flow will help your team be more productive at the desk. Contact your Logitech sales representative to better understand how a Logitech Flow-enabled mouse could help your team be more productive at a desk.

ABOUT LOGITECH

Logitech designs products that have an everyday place in people's lives, connecting them to the digital experiences they care about. More than 35 years ago, Logitech started connecting people through computers, and now it's a multi-brand company designing products that bring people together through music, gaming, video and computing. Brands of Logitech include Jaybird, Logitech G and Ultimate Ears. Founded in 1981, and headquartered in Lausanne, Switzerland, Logitech International is a Swiss public company listed on the SIX Swiss Exchange (LOGN) and on the Nasdaq Global Select Market (LOGI). Find Logitech at www.logitech.com, the company blog or @Logitech.